The Science of Monetary Policy: A New Keynesian Perspective

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Key concepts

- Euler equation, Fisher equation, dynamic IS curve
- Dixit-Stiglitz aggregator, Calvo pricing, New Keynesian Phillips Curve (NKPC)
- discretion vs commitment
- optimal monetary policy, divine coincidence
- monetary policy rules, implementability

Household block

Household problem

Household maximizes his/her utility subject to budget constraint:

$$\max \sum_{t=0}^{\infty} \beta^t U(C_t, \ell_t)$$

$$P_t C_t + Q_t B_t = B_{t-1} + W_t \ell_t + Div_t$$

Where $Q_t = \frac{1}{R_t} = \frac{1}{1+i_t}$.
Typically we assume:

$$U(C_t, \ell_t) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\ell_t^{1+\varphi}}{1+\varphi}$$

Solving HH problem

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\ell_t^{1+\varphi}}{1+\varphi} + \right.$$
$$\left. + \lambda_t [B_{t-1} + W_t \ell_t + Div_t - P_t C_t - Q_t B_t] \right\}$$

First order conditions:

$$C_t: \quad \beta^t C_t^{-\sigma} = \beta^t \lambda P_t$$
$$\ell_t: \quad \beta^t \ell_t^{\varphi} = \beta^t \lambda W_t$$
$$B_t: \quad \beta^t Q_t \lambda_t = \beta^{t+1} \mathbb{E}_t \lambda_{t+1}$$

First order conditions

$$\ell_t^{\varphi} C_t^{-\sigma} = \frac{W_t}{P_t} = w_t \qquad \text{(labor supply)}$$

$$Q_t = \beta \mathbb{E}_t \left\{ \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} \right\} \qquad \text{(Euler equation)}$$

Dynamic IS curve:

$$x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\sigma}[i_t - \mathbb{E}_t \pi_{t+1}] + g_t$$

Firm block

Final goods producer

Final goods firm produces according to CES production function (Dixit-Stiglitz aggregator):

$$Y_t = \left(\int_0^1 Y_t^{\frac{\phi-1}{\phi}}(i)di \right)^{\frac{\phi}{\phi-1}}$$

$Y_t(i)$ is output of intermediate good produced by firm i, $i \in [0,1]$ continuum of firms

Demand for good produced by firm i:

$$Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\phi} Y_t$$

Intermediate firm price setting problem

Monopolistic competition.

If prices are flexible (no price stickiness due to nominal rigidities):

$$\max_{P_t(i)} \left(P_t(i) - \frac{W_t}{A_t} \right) \left(\frac{P_t(i)}{P_t} \right)^{-\phi} Y_t$$

from that we get:

$$P_t(i) = \frac{\phi}{\phi - 1} \underbrace{\frac{W_t}{A_t}}_{MC_t} > MC_t$$

Nominal rigidities

There are different ways to introduce nominal rigidities:

1. time dependent pricing:
   - Calvo pricing (Calvo 1983), every period random fraction θ cannot change price
   - Taylor contracts (Taylor 1980), firm can change price every T periods
2. state dependent pricing:
   - Rotemberg pricing (Rotemberg 1982)
   - menu-cost models (Golosov, Lucas 2007)
3. imperfect information:
   - sticky information (Mankiw, Reis 2002)

Price setting with Calvo pricing

Maximization problem of firm i under sticky prices:

$$\max_{\tilde{P}_t} \left\{ \mathbb{E}_t \sum_{i=0}^{\infty} \Lambda_{t,t+i} \beta^i \theta^i \left(\tilde{P}_t - \frac{W_{t+i}}{A_{t+i}} \right) \left(\frac{\tilde{P}_t}{P_t} \right)^{-\phi} Y_t \right\}$$

From which we get New Keynesian Phillips curve (NKPC):

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \lambda x_t + u_t$$

Monetary policy

Forward iteration

We have 2/3 of the famous equations:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \lambda x_t + u_t \qquad \text{(NKPC)}$$

$$x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\sigma}[i_t - \mathbb{E}_t \pi_{t+1}] + g_t \qquad \text{(IS curve)}$$

For intuition we can iterate them forward:

$$\pi_t = \mathbb{E}_t \sum_{i=0}^{\infty} \left[\beta^i (\lambda x_{t+i} + u_{t+i})\right]$$

$$x_t = \mathbb{E}_t \sum_{j=0}^{\infty} \left[-\frac{1}{\sigma}(i_{t+j} - \pi_{t+j+1}) + g_{t+j}\right]$$

Central bank objective

$$\max -\frac{1}{2}\mathbb{E}_t \left\{ \sum_{j=0}^{\infty} \beta^j [\alpha x_{t+i}^2 + \pi_{t+i}^2] \right\}$$

Discretion vs commitment

Discretion:

$$\pi_{t+j} = -\frac{\alpha}{\lambda} x_t$$

Commitment:

$$\pi_{t+j} = -\frac{\alpha}{\lambda}(x_{t+j} - x_{t+j-1})$$

Result 1:

To the extent cost push inflation is present, there exists a short run trade-off between inflation and output variability

Discretion vs commitment II

Inflationary bias problem:

$$\max -\frac{1}{2}\mathbb{E}_t \left\{ \sum_{j=0}^{\infty} \beta^j [\alpha(x_{t+i} - k)^2 + \pi_{t+i}^2] \right\}$$

Solution:

$$x_t^k = x_t$$
$$\pi_t^k = \pi_t + \frac{\alpha}{\lambda} k$$

Result 5:

If the central bank desires to push output above potential (i.e., $k > 0$), then under discretion a suboptimal equilibrium may emerge with inflation persistently above target, and no gain in output.

Monetary policy rules

$$i_t = r_t^n + \phi_\pi \pi_t + \phi_x x_t$$

Taylor rule

$i_t = \rho + \phi_\pi \pi_t + \phi_x x_t$

e.g. in Taylor (1993): $i_t = 2\% + \pi_t + 0.5(\pi_t - \bar{\pi}) + 0.5 x_t$

Result 3 (Taylor principle):

Under the optimal policy, in response to a rise in expected inflation, nominal rates should rise sufficiently to increase real rates. Put differently, in the optimal rule for the nominal rate, the coefficient on expected inflation should exceed unity.

3-equation NK model

$$x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\sigma}\left(R_t - \mathbb{E}_t\{\pi_{t+1}\} - r_t^n\right) \quad \text{(dynamic IS curve)}$$

$$\pi_t = \beta \mathbb{E}_t\{\pi_{t+1}\} + \lambda x_t \quad \text{(Phillips curve)}$$

$$R_t = r_t^n + \phi_\pi \pi_t + \phi_x x_t \quad \text{(Taylor rule)}$$

3-equation NK model II

$$\phi_\pi \pi_t + (\sigma + \phi_x) x_t = \sigma \mathbb{E}_t\{x_{t+1}\} + \mathbb{E}_t\{\pi_{t+1}\}$$
$$\pi_t - \lambda x_t = \beta \mathbb{E}_t\{\pi_{t+1}\}$$

In matrix form:

$$\begin{bmatrix} \mathbb{E}_t x_{t+1} \\ \mathbb{E}_t \pi_{t+1} \end{bmatrix} = \frac{1}{\sigma\beta} \begin{bmatrix} \beta(\sigma + \phi_x) + \lambda & \beta\phi_\pi - 1 \\ -\lambda\sigma & \sigma \end{bmatrix} \begin{bmatrix} x_t \\ \pi_t \end{bmatrix}$$

